吉林大学

DB2 实验报告

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姓名	柳湛博	学号	55241019
实验项目	3. Create objects		
实验性质	☐演示性实验 ☐验证性实验 ☑操作性实验 ☐综合性实验		
实验地点	A108	机器编号	10
指导教师	刘雪洁	实验时间	2026 年 9 月 25 日 5 时 45 分

一、实验综述

1. 实验目的及要求

This exercise is an online lab which creates objects used in a database.

At the end of the lab, students should be able to:

- Create tables
- Create indexes
- Create views
- Create an alias
- Add referential integrity constraints to a table
- Add check constraints to a table
- Add triggers to a table
- Access System Catalog information about objects
- Retrieve an XML document

2. 实验设备、软件

PC, windows XP Professional, DB2 9 Express-c

二、实验过程（实验步骤、记录、数据、分析）

Section 1 - Create Tables

1. Create your ARTISTS table. The table should have the following characteristics.
   - Table schema of student
   - Table Name of artists
   - Regular data to go in table space dms01
   - Indexes to go in table space dms02
   - Long data to go in table space dms03
   - The column artno should be defined as a primary key
   - Columns defined like the following:

```
( artno smallint not null,
  name varchar (50),
  classification char (1) not null,
  bio clob (100K) logged compact,
picture blob (500k) not logged compact)
```

Show your work below by printing screen.



2. Verify the creation of your ARTISTS table and the table description.
Important: Ensure the definition of your table is correct. It could cause unexpected failures in later labs if your definitions are not done correctly.
Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 describe table artists show deta
il

                                                                                列
                列          分区键          代码
列名
                                                数据类型模式

                                数据类型名称
 长          小数位     NULL    号          序列                  页        缺省值
------------------------------------------------------------------------------
------------------------------------------- -------------------------------
------------------------------------------------------------------------------
--------------- ------------------------------------------------------------
------------------------------------------------------------ ---------- -
------- -------- ---------- ------------------ -------- --------------------
-----------------
ARTNO
                                        SYSIBM

                SMALLINT
                                                                          2
     0 否               0                  0        0

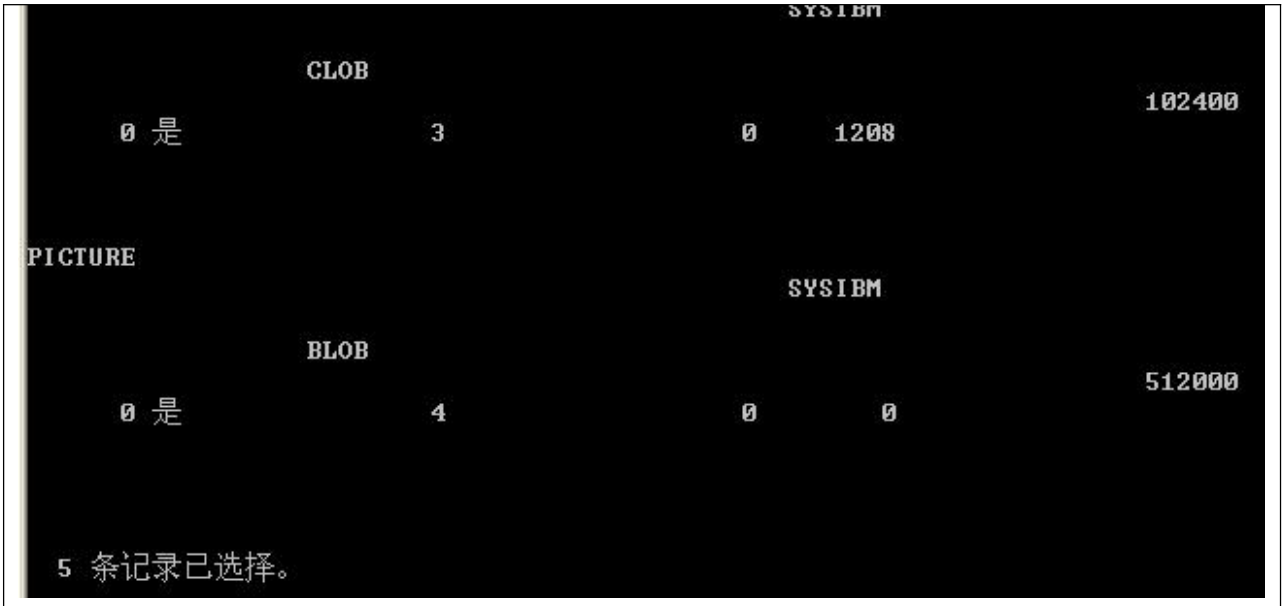

NAME
                                        SYSIBM

                VARCHAR
                                                                         50
     0 是               1                  0     1208


CLASSIFICATION
                                        SYSIBM

                CHARACTER
                                                                          1
     0 否               2                  0     1208


BIO
                                        SYSIBM
```



3. You will create the remainder of the tables required using a script file called crtables. For Windows, it is located in the C:\labfiles\cf23 directory.

Examine the DB2 statements in the file and answer the following questions.

In which table space will both the indexes and data for table STOCK be placed? For CONCERTS? For REORDER?

STOCK 的数据和索引均位于 DMS06；CONCERTS 的数据和索引均位于 DMS04；REORDER 的数据和索引均位于 SMS01。

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>type CRTABLES
connect to musicdb;

create table albums
  (title     varchar (50),
   artno     smallint not null,
   itemno    smallint not null)
         in dms04
   index in dms05;

create table stock
 (itemno             smallint not null,
  type               char (1) not null,
  price              decimal (5,2) not null with
default,
  qty                int not null with default)
         in dms06;

create table concerts
 (artno              smallint not null,
  date               date not null,
  city               varchar (25) not null with
default)
         in dms04;

create table reorder
  (itemno           smallint  not null,
   timestamp        timestamp)
         in sms01;
```

4. Execute the crtables script to create the tables.
Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tvf CRTABLES
connect to lzb_1019

   数据库连接信息

 数据库服务器         = DB2/NT 9.7.0
 SQL 授权标识         = LZB_1019
 本地数据库别名       = LZB_1019


create table albums (title     varchar (50), artno     smallint not null, itemno
    smallint not null) in dms04 index in dms05
DB20000I  SQL 命令成功完成。

create table stock (itemno             smallint not null, type               cha
r (1) not null, price              decimal (5,2) not null with default, qty
          int not null with default) in dms06
DB20000I  SQL 命令成功完成。

create table concerts (artno              smallint not null, date
date not null, city              varchar (25) not null with default) in dms04
DB20000I  SQL 命令成功完成。

create table reorder (itemno           smallint  not null, timestamp        time
stamp) in sms01
DB20000I  SQL 命令成功完成。
```

5. Verify that all tables (ALBUMS, STOCK, CONCERTS, and REORDER) were created successfully.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "select substr(tabname,1,18) as
tabname, type from syscat.tables where tabschema=user and tabname in ('ALBUMS','
ARTISTS','STOCK','CONCERTS','REORDER') order by tabname"

TABNAME            TYPE
------------------ ----
ALBUMS             T
ARTISTS            T
CONCERTS           T
REORDER            T
STOCK              T

  5 条记录已选择。
```

6. Information about each column in a table can be found by accessing the SYSCAT.COLUMNS view.

Enter an interactive SQL statement to select the columns TABNAME, COLNAME, and TYPENAME from SYSCAT.COLUMNS for all tables with the TABSCHEMA of 'student', order the data by TABNAME and COLNO, and pipe the output to a file named tabchk.file.

Show your sql statements and your work below by printing screen for the last step.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tf tabchk.sql > tabchk.file

D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>type tabchk.file

TABNAME
                                                  COLNAME

                    TYPENAME

------------------------------------------------------------------------------
------------------------------------------------ ----------------------------
------------------------------------------------------------------------------
---------------- ------------------------------------------------------------
------------------------------------------------------------

ALBUMS
                                                  ARTNO

                    SMALLINT

ALBUMS
                                                  ITEMNO

                    SMALLINT

ALBUMS
                                                  TITLE

                    VARCHAR

ARTISTS
                                                  ARTNO

                    SMALLINT

ARTISTS
                                                  BIO

                    CLOB

ARTISTS
                                                  CLASSIFICATION

                    CHARACTER

ARTISTS
                                                  NAME

                    VARCHAR

ARTISTS
                                                  PICTURE

                    BLOB                                                D:\DB2\LabFil
```

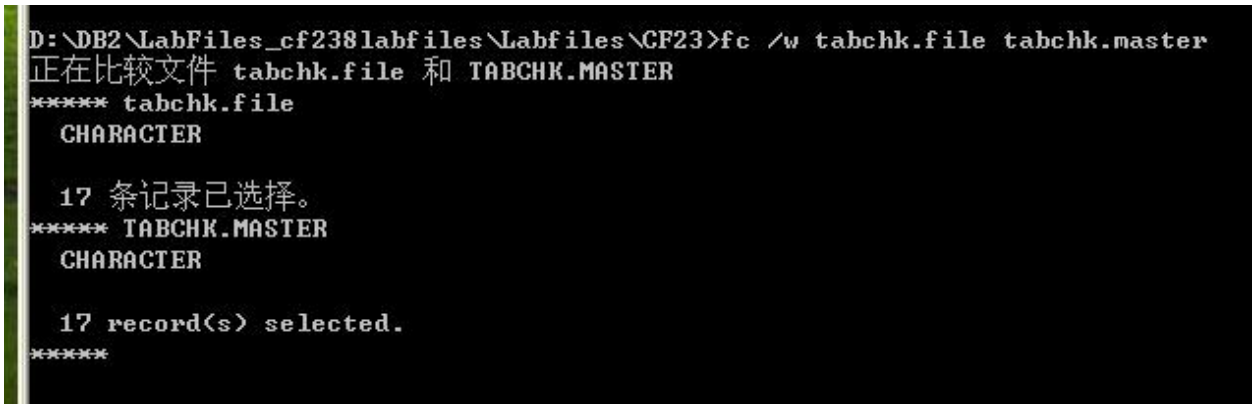
```
CONCERTS                                                ARTNO
                SMALLINT
CONCERTS                                                CITY
                VARCHAR
CONCERTS                                                DATE
                DATE
REORDER                                                 ITEMNO
                SMALLINT
REORDER                                                 TIMESTAMP
                TIMESTAMP
STOCK                                                   ITEMNO
                SMALLINT
STOCK                                                   PRICE
                DECIMAL
STOCK                                                   QTY
                INTEGER
STOCK                                                   TYPE
                CHARACTER

  17 条记录已选择。
```

7. Check your output against the tabchk.master file.

Ensure that there are no differences between your output and the master output or the following labs may not work correctly!

Show your work below by printing screen.



除结尾系统语言外无差别

8. Information about a table's definition to a table space can be found by accessing the SYSCAT.TABLES view. Enter an SQL statement to select the columns TABNAME, TBSPACE, and INDEX_TBSPACE from SYSCAT.TABLES for all tables with the TABSCHEMA of ‘student’, order the data by TABNAME, and pipe the output to a file named tbschk.file.

Show your sql statements and your work below by printing screen for the last step.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tf tbschk.sql > tbschk.file

D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>type tbschk.file

TABNAME
                                                TBSPACE

                  INDEX_TBSPACE

------------------------------------------------------------------------------
------------------------------------------------ --------------------------------
------------------------------------------------------------------------------
------------------ --------------------------------------------------------------
------------------------------------------------------------------

ALBUMS
                                                DMS04

                  DMS05

ARTISTS
                                                DMS01

                  DMS02

CONCERTS
                                                DMS04

                  -

REORDER
                                                SMS01

                  -

STOCK
                                                DMS06

                  -


  5 条记录已选择。
```

9. Check your output against the tbschk.master file.

Ensure that there are no differences between your output and the master output, or the following labs may not work correctly!

Show your work below by printing screen.



除结尾系统语言外无差别

10. Grant select privilege on tables that are owned by student to public by executing a script file, grants (an operating system command file) for local administration.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tvf GRANTS
connect to lzb_1019

   数据库连接信息

 数据库服务器          = DB2/NT 9.7.0
 SQL 授权标识          = LZB_1019
 本地数据库别名        = LZB_1019


grant select on table artists to public
DB20000I  SQL 命令成功完成。

grant select on table albums to public
DB20000I  SQL 命令成功完成。

grant select on table stock to public
DB20000I  SQL 命令成功完成。

grant select on table concerts to public
DB20000I  SQL 命令成功完成。

grant select on table reorder to public
DB20000I  SQL 命令成功完成。
```

Section 2 - Create Indexes

1. Create an index called ITEM on the ITEMNO column in STOCK table.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "create index item on stock (ite
mno)"
DB20000I  SQL 命令成功完成。
```

2. Create a unique index called ITEMNO on the ITEMNO column in ALBUMS table.

Show your work below by printing screen.



3. Select the information from the catalog tables about your indexes. The columns you should select are the first 18 characters of TABNAME, UNIQUERULE, the first 18 characters of INDNAME, and the first 30 characters of COLNAMES from SYSCAT.INDEXES, where INDSCHEMA is student, order by TABNAME and INDNAME.

If UNIQUERULE = "U", then only unique values are allowed.

If UNIQUERULE = "D", then duplicate values are allowed.

If UNIQUERULE = "P", then it is a Primary key.

Show your work and your new indexes below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "select substr(tabname,1,18) as
tabname, uniquerule, substr(indname,1,18) as indname, substr(colnames,1,30) as c
olnames from syscat.indexes where indschema=user order by tabname, indname"

TABNAME            UNIQUERULE INDNAME            COLNAMES
------------------ ---------- ------------------ ------------------------------
ALBUMS             U          ITEMNO             +ITEMNO
STOCK              D          ITEM               +ITEMNO

  2 条记录已选择。
```

4. Use the select you executed in the previous step and route the output to indchk.file. Check your results against indchk.master.

Ensure that there are no differences between your output and the master output, or the following labs may not work correctly!

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>fc /w indchk.file indchk.master
正在比较文件 indchk.file 和 INDCHK.MASTER
***** indchk.file
STOCK              D          ITEM               +ITEMNO
  2 条记录已选择。
***** INDCHK.MASTER
STOCK              D          ITEM               +ITEMNO
  2 record(s) selected.
*****
```

除结尾语言外无差异

Section 3 - Create Views

1. Create a view called music that will select title, classification, name from albums, artists where artists.artno = albums.artno.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "create view music as select alb
ums.title, artists.classification, artists.name from albums, artists where artis
ts.artno = albums.artno"
DB20000I  SQL 命令成功完成。
```

2. Create another view called inventory by executing the script file, crview.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tvf CRVIEW
connect to lzb_1019

   数据库连接信息

 数据库服务器          = DB2/NT 9.7.0
 SQL 授权标识          = LZB_1019
 本地数据库别名        = LZB_1019


create view inventory (type, itemno, totcost, totqty) as select type, itemno, su
m (price * qty), sum(qty) from stock group by type, itemno
DB20000I  SQL 命令成功完成。
```

3. Information about views can be found in the SYSCAT.VIEWS and SYSCAT.TABLES views. Tables and views for the current user can be shown with a LIST TABLES statement. A TYPE of V is a view.

Issue one of the following statements and verify your views are listed.

- LIST TABLES statement OR
- Select columns TABSCHEMA, TABNAME, and TYPE from the SYSCAT.TABLES view with a TABSCHEMA of student OR
- Select columns VIEWSCHEMA, VIEWNAME from SYSCAT.VIEWS with DEFINER of student

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "select substr(tabname,1,18) as
tabname, type from syscat.tables where tabschema=user and type='V' order by tabn
ame"

TABNAME            TYPE
------------------ ----
INVENTORY          V
MUSIC              V

  2 条记录已选择。
```

Section 4 - Create Alias

1. Create an alias called singers for the ARTISTS table, and an alias called emptystock for the REORDER table.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "create alias singers for artist
s"
DB20000I  SQL 命令成功完成。

D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "create alias emptystock for reo
rder"
DB20000I  SQL 命令成功完成。
```

2. Information about aliases can be found in the SYSCAT.TABLES view. Tables, views, and aliases for the current user can also be shown with the LIST TABLES statement. A TYPE of A is an alias. Issue one of the following statements and verify your aliases are listed.

- LIST TABLES statement OR
- Select columns TABNAME and TYPE from the SYSCAT.TABLES view with a TABSCHEMA of student.

Show your work below by printing screen.



Section 5 - Add Referential Integrity

1. Alter the ALBUMS table and define referential integrity relationships it has with other tables.

- Add a primary key on the itemno column
- Add a foreign key called fkartno on the artno column which references a primary key in the ARTISTS table
- The delete rule between ARTISTS and ALBUMS should be delete cascade

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "alter table albums add primary
key (itemno) add constraint fkartno foreign key (artno) references artists (artn
o) on delete cascade"
DB20000I  SQL 命令成功完成。

D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>
```

2. Now alter the STOCK table to define its referential integrity relationships, by executing the script file, crri.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tvf CRRI
connect to lzb_1019

   数据库连接信息

 数据库服务器          = DB2/NT 9.7.0
 SQL 授权标识          = LZB_1019
 本地数据库别名        = LZB_1019


alter table stock foreign key fkitemno (itemno) references albums on delete rest
rict
DB20000I  SQL 命令成功完成。
```

3. Information about referential integrity can be found by accessing the SYSCAT.REFERENCES view. Table dependencies can also be accessed through the SYSCAT.TABLES view. Enter an interactive SQL statement to select CONSTNAME, TABNAME, REFTABSCHEMA, REFTABNAME, DELETERULE from the SYSCAT.REFERENCES view with a TABSCHEMA of student.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "select substr(constname,1,18) a
s constname, substr(tabname,1,18) as tabname, substr(reftabname,1,18) as reftabn
ame, deleterule from syscat.references where tabschema=user order by tabname, co
nstname"

CONSTNAME          TABNAME            REFTABNAME         DELETERULE
------------------ ------------------ ------------------ ----------
FKARTNO            ALBUMS             ARTISTS            C
FKITEMNO           STOCK              ALBUMS             R

  2 条记录已选择。
```

4. Run a select to retrieve the first 18 characters of TABNAME, and the PARENTS and CHILDREN columns from SYSCAT.TABLES where the TABSCHEMA is equal to your userid. Order the results by TABNAME and direct your output to a file named richk.file. The SQL can be found in a file named richk.sql.

Check your output against the richk.master file.

Ensure that there are no differences between your output and the master output, or the following labs may not work correctly!

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tf richk.sql > richk.file

D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>type richk.file

1                    PARENTS CHILDREN
-------------------- ------- --------
ALBUMS                     1        1
ARTISTS                    0        1
CONCERTS                   0        0
EMPTYSTOCK                 0        0
INVENTORY                  0        0
MUSIC                      0        0
REORDER                    0        0
SINGERS                    0        0
STOCK                      1        0

  9 条记录已选择。



D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>fc /w richk.file richk.master
正在比较文件 richk.file 和 RICHK.MASTER
***** richk.file
STOCK                      1        0
  9 条记录已选择。
***** RICHK.MASTER
STOCK                      1        0
  9 record(s) selected.
*****
```

除结尾语言外无差异

Section 6 - Add Check Constraints

1. Alter the STOCK table and add a check constraint to it. The check constraint should have the following characteristics.

- It should be named cctype.
- The business rule it should enforce is to only allow the values of 'D', 'C', or 'R' in the type column.

Make sure the alphanumeric literals are typed in uppercase letters.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "alter table stock add constrain
t cctype check (type in ('D', 'C', 'R'))"
DB20000I  SQL 命令成功完成。
```

2. Information about check constraints can be found by accessing the SYSCAT.CHECKS,SYSCAT.COLCHECKS, SYSCAT.TABCONST, and SYSCAT.TABLES views. Issue SQL to:

Select CONSTNAME, TABNAME, COLNAME from the SYSCAT.COLCHECKS view.

Select CONSTNAME, TABNAME, TYPE from the SYSCAT.TABCONST view.

If the TYPE = “K”, then it is a check constraint. If the Type = “P”, then it is a primary key. If the Type = “F”, then it is a foreign key.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "select substr(constname,1,18) a
s constname, substr(tabname,1,18) as tabname, substr(colname,1,18) as colname fr
om syscat.colchecks where tabschema=user order by tabname, constname"

CONSTNAME          TABNAME            COLNAME
------------------ ------------------ ------------------
CCTYPE             STOCK              TYPE

  1 条记录已选择。


D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "select substr(constname,1,18) a
s constname, substr(tabname,1,18) as tabname, type from syscat.tabconst where ta
bschema=user order by tabname, type, constname"

CONSTNAME          TABNAME            TYPE
------------------ ------------------ ----
FKARTNO            ALBUMS             F
SQL261010200719770 ALBUMS             P
SQL261010195222600 ARTISTS            P
FKITEMNO           STOCK              F
CCTYPE             STOCK              K

  5 条记录已选择。
```

3. To check your work, run a select statement that selects the first 100 characters of the TEXT column from SYSCAT.CHECKS where CONSTNAME = 'CCTYPE', and direct your output to a file named ckchk.file. The SQL can be found in a file named ckchk.sql. Check your output against the ckchk.master file.

Ensure that there are no differences between your output and the master output, or the following labs may not work correctly!

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tf ckchk.sql > ckchk.file

D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>type ckchk.file

1

--------------------------------------------------------------------------------
--------------------
TYPE in ('D','C','R')


  1 条记录已选择。



D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>fc /w ckchk.file ckchk.master
正在比较文件 ckchk.file 和 CKCHK.MASTER
***** ckchk.file
TYPE in ('D','C','R')

  1 条记录已选择。
***** CKCHK.MASTER
TYPE in ('D','C','R')

  1 record(s) selected.
*****
```

除结尾语言外无差别

Section 7 - Create a Trigger

1. Create a trigger which has the following characteristics.

- It should be named reorder
- It should fire after an update of the qty column on the STOCK table, if the new value of qty is <= 5
- New should be referenced as n
- The triggered action should insert the values n.itemno and current timestamp into the REORDER table

• For each row mode db2sql

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "create trigger reorder after up
date of qty on stock referencing new as n for each row mode db2sql when (n.qty <
= 5) insert into reorder values (n.itemno, current timestamp)"
DB20000I  SQL 命令成功完成。
```

2. Information about triggers can be found by accessing the SYSCAT.TRIGGERS and SYSCAT.TRIGDEP views.

Enter interactive SQL to research the REORDER trigger:

Select TRIGNAME, TABNAME, and TRIGEVENT columns from the SYSCAT.TRIGGERS view.

TRIGEVENT describes the event that fires the trigger as I for insert, D for delete, or U for update.

Select TRIGNAME, BTYPE, BSCHEMA, BNAME columns from SYSCAT.TRIGDEP view. BTYPE and BSCHEMA give the name of the objected depended on by the trigger. BTYPE describes the type of base object as A for alias, F for function instance, T for table, or V for view.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "select substr(trigname,1,18) as
 trigname, substr(tabname,1,18) as tabname, trigevent from syscat.triggers where
 trigschema=user order by trigname"

TRIGNAME           TABNAME            TRIGEVENT
------------------ ------------------ ---------
REORDER            STOCK              U

  1 条记录已选择。


D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "select substr(trigname,1,18) as
 trigname, btype, substr(bschema,1,12) as bschema, substr(bname,1,18) as bname f
rom syscat.trigdep where trigschema=user and trigname='REORDER' order by btype,
bname"

TRIGNAME           BTYPE BSCHEMA      BNAME
------------------ ----- ------------ ------------------
REORDER            T     LZB_1019     REORDER
REORDER            T     LZB_1019     STOCK

  2 条记录已选择。
```

Section 8 - Working with XML

1. If the SAMPLE database has not yet been create, create it now.

Show your work below by printing screen.

已有 Sample 数据库

2. Connect to the SAMPLE database.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 connect reset
DB20000I  SQL 命令成功完成。

D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 connect to sample

   数据库连接信息

 数据库服务器          = DB2/NT 9.7.0
 SQL 授权标识          = LZB_1019
 本地数据库别名        = SAMPLE
```

3. Use XQUERY and the db2-fn:xmlcolumn function to retrieve all of the XML documents from the customer table's info column.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "xquery db2-fn:xmlcolumn('ADMINI
STRATOR.CUSTOMER.INFO')"
```

```
<customerinfo Cid="1000"><name>Kathy Smith</name><addr country="Canada"><street>
5 Rosewood</street><city>Toronto</city><prov-state>Ontario</prov-state><pcode-zi
p>M6W 1E6</pcode-zip></addr><phone type="work">416-555-1358</phone></customerinf
o>
```

```
<customerinfo Cid="1001"><name>Kathy Smith</name><addr country="Canada"><street>
25 EastCreek</street><city>Markham</city><prov-state>Ontario</prov-state><pcode-
zip>N9C 3T6</pcode-zip></addr><phone type="work">905-555-7258</phone></customeri
nfo>
```

```
<customerinfo Cid="1002"><name>Jim Noodle</name><addr country="Canada"><street>2
5 EastCreek</street><city>Markham</city><prov-state>Ontario</prov-state><pcode-z
ip>N9C 3T6</pcode-zip></addr><phone type="work">905-555-7258</phone></customerin
fo>
```

```
<customerinfo Cid="1003"><name>Robert Shoemaker</name><addr country="Canada"><st
reet>1596 Baseline</street><city>Aurora</city><prov-state>Ontario</prov-state><p
code-zip>N8X 7F8</pcode-zip></addr><phone type="work">905-555-7258</phone><phone
 type="home">416-555-2937</phone><phone type="cell">905-555-8743</phone><phone t
ype="cottage">613-555-3278</phone></customerinfo>
```

```
<customerinfo Cid="1004"><name>Matt Foreman</name><addr country="Canada"><street
>1596 Baseline</street><city>Toronto</city><prov-state>Ontario</prov-state><pcod
e-zip>M3Z 5H9</pcode-zip></addr><phone type="work">905-555-4789</phone><phone ty
pe="home">416-555-3376</phone><assistant><name>Gopher Runner</name><phone type="
home">416-555-3426</phone></assistant></customerinfo>
```

```
<customerinfo Cid="1005"><name>Larry Menard</name><addr country="Canada"><street
>223 NatureValley Road</street><city>Toronto</city><prov-state>Ontario</prov-sta
te><pcode-zip>M4C 5K8</pcode-zip></addr><phone type="work">905-555-9146</phone><
phone type="home">416-555-6121</phone><assistant><name>Goose Defender</name><pho
ne type="home">416-555-1943</phone></assistant></customerinfo>
```

```
6 条记录已选择。
```

4. Now use XQUERY with SQL and db2-fn:sqlquery function to retrieve all of the XML documents from the customer table's info column.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -x "xquery db2-fn:sqlquery('sele
ct info from ADMINISTRATOR.customer')" > customer_info_8_4.txt

D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>more customer_info_8_4.txt
<customerinfo Cid="1000"><name>Kathy Smith</name><addr country="Canada"><street>
5 Rosewood</street><city>Toronto</city><prov-state>Ontario</prov-state><pcode-zi
p>M6W 1E6</pcode-zip></addr><phone type="work">416-555-1358</phone></customerinf
o>
```

```
<customerinfo Cid="1001"><name>Kathy Smith</name><addr country="Canada"><street>
25 EastCreek</street><city>Markham</city><prov-state>Ontario</prov-state><pcode-
zip>N9C 3T6</pcode-zip></addr><phone type="work">905-555-7258</phone></customeri
nfo>
```

```
<customerinfo Cid="1002"><name>Jim Noodle</name><addr country="Canada"><street>2
5 EastCreek</street><city>Markham</city><prov-state>Ontario</prov-state><pcode-z
ip>N9C 3T6</pcode-zip></addr><phone type="work">905-555-7258</phone></customerin
fo>
```

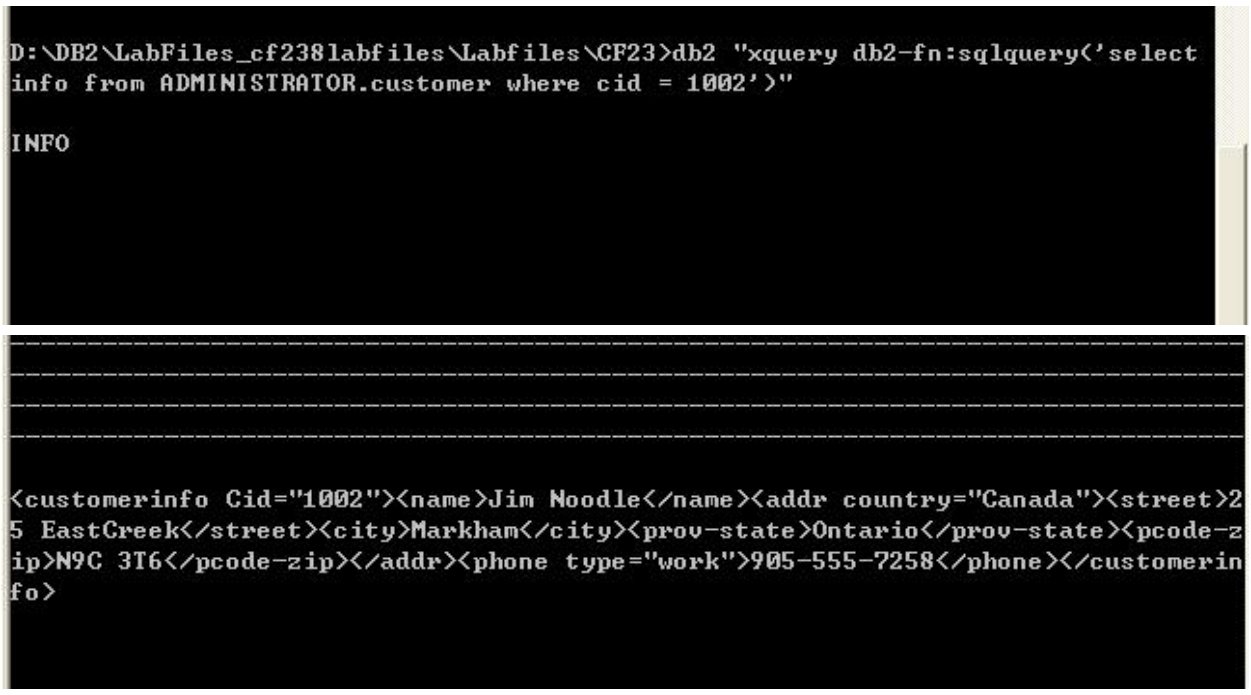
```
<customerinfo Cid="1003"><name>Robert Shoemaker</name><addr country="Canada"><st
reet>1596 Baseline</street><city>Aurora</city><prov-state>Ontario</prov-state><p
code-zip>N8X 7F8</pcode-zip></addr><phone type="work">905-555-7258</phone><phone
 type="home">416-555-2937</phone><phone type="cell">905-555-8743</phone><phone t
ype="cottage">613-555-3278</phone></customerinfo>
```

```
<customerinfo Cid="1004"><name>Matt Foreman</name><addr country="Canada"><street
>1596 Baseline</street><city>Toronto</city><prov-state>Ontario</prov-state><pcod
e-zip>M3Z 5H9</pcode-zip></addr><phone type="work">905-555-4789</phone><phone ty
pe="home">416-555-3376</phone><assistant><name>Gopher Runner</name><phone type="
home">416-555-3426</phone></assistant></customerinfo>
```

```
<customerinfo Cid="1005"><name>Larry Menard</name><addr country="Canada"><street
>223 NatureValley Road</street><city>Toronto</city><prov-state>Ontario</prov-sta
te><pcode-zip>M4C 5K8</pcode-zip></addr><phone type="work">905-555-9146</phone><
phone type="home">416-555-6121</phone><assistant><name>Goose Defender</name><pho
ne type="home">416-555-1943</phone></assistant></customerinfo>
```

5. Use XQUERY and the db2-fn:sqlquery function to retrieve, from the INFO column in the CUSTOMER table, all customers where the customer ID (column CID) equals 1002.

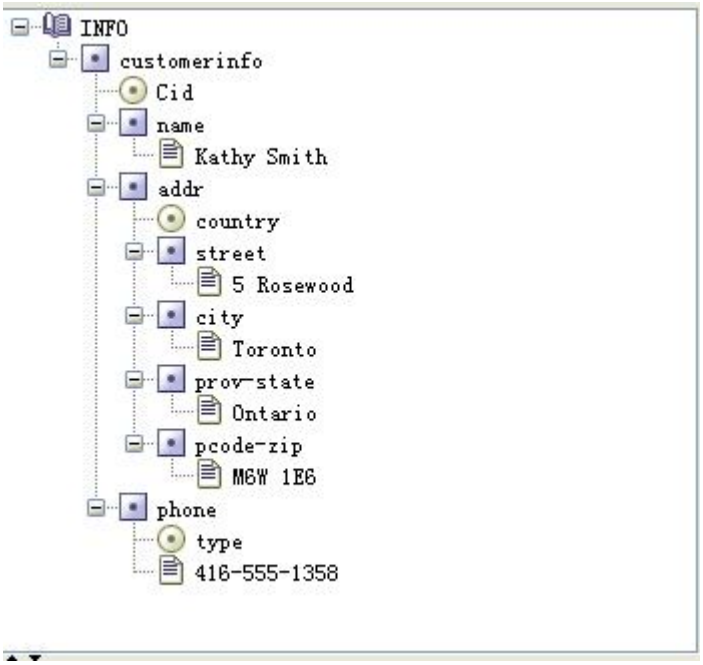
Show your work below by printing screen.



6. Use the DB2 Control Center and the XML Document View to display the CUSTOMER.INFO document in the first row of the table. View the document in

both Tree view and Source view.

Show your work below by printing screen.




```
<customerinfo Cid="1000"><name>Kathy Smith</name><addr country="Canada">
><street>5 Rosewood</street><city>Toronto</city><prov-state>Ontario
</prov-state><pcode-zip>M6W 1E6</pcode-zip></addr><phone type="work">
.e-zip></addr><phone type="work">416-555-1358</phone></customerinfo>
```